

MSc

Course title: SM6 Bayes Methods
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1. For $i = 1, \dots, n$ let $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})$, $\mathbf{x}_i \in \mathbb{R}^p$ be a vector of p continuous covariates for the i 'th observation. Let $\mathcal{M}$ be the set of all subsets of $\{1, 2, \dots, p\}$ (including the empty set) and let $m \in \mathcal{M}$ be one such set. Let $\beta = (\beta_1, \dots, \beta_p)$, $\beta \in \mathbb{R}^p$ be a vector of regression parameters. For $m \in \mathcal{M}$, let $K = |m|$ give the number of elements in m . The labels in m are ordered so if $m = \{m_1, \dots, m_K\}$ then $m_1 < m_2 < \dots < m_K$. Denote by $\beta_m = (\beta_{m_1}, \dots, \beta_{m_K})$ the vector of parameters in the model when $|m| = K$. For $m \in \mathcal{M}$ with $K = |m|$, let

$$y_i = \sum_{j=1}^K \beta_{m_j} x_{i,m_j} + \epsilon_i,$$

with $\epsilon_i \sim N(0, \sigma^2)$ jointly independent for $i = 1, \dots, n$, be the observation model for $y = (y_1, \dots, y_n)$ given the unknown true model is model $M = m$. Here $y_i = \epsilon_i$, $i = 1, \dots, n$ if $m = \emptyset$ so $K = 0$. Here $(\beta_m, \sigma) \in \Omega_m$ where $\Omega_m = \mathbb{R}^K \times (0, \infty)$ and $K = |m|$. A prior

$$\pi(\beta_m, \sigma | m) = \pi_\sigma(\sigma) \prod_{j=1}^K \pi(\beta_{m_j})$$

is given. The prior $\pi(\beta_m, \sigma | m)$ is a probability density on Ω_m . A model prior probability mass function $\pi_M(m)$, $m \in \mathcal{M}$ is given.

(a) [10 marks]

- (i) Write down the fixed-model posterior $\pi(\beta_m, \sigma | y, m)$ in terms of the normal observation model density and priors.
- (ii) Give the joint posterior distribution for the model and parameter, $\pi(\beta_m, \sigma, m | y)$, in terms of the model elements, and the state space for (β_m, σ, m) .
- (iii) Let model $m' = \{1, \dots, p\}$ be the full model and suppose that, for $m \in \mathcal{M}$, estimates $\hat{B}_{m,m'}$ of ratios $B_{m,m'} = p(y|m)/p(y|m')$ are available. For $m \in \mathcal{M}$ and $t = 1, \dots, T$, let $\beta^{(m,t)}, \sigma^{(m,t)}$, $t = 1, \dots, T$ be samples distributed according to $\pi(\beta_m, \sigma | y, m)$. Let $g(\beta_m, \sigma, m)$ be a given function of the parameters and model index. Give an estimator for the model-averaged posterior expectation $E_{\beta_m, \sigma, m | y}(g(\beta_m, \sigma, m))$ in terms of $\hat{B}_{m,m'}$, $m \in \mathcal{M}$ and the samples.

(b) [10 marks]

- (i) For $c > 0$ let $(\rho_{m,m'})_{m,m' \in \mathcal{M}}$ be a matrix satisfying $\rho_{m,m'} = c$ if m and m' differ by exactly one element (and otherwise equal zero), so if

$$m \Delta m' = (m \setminus m') \cup (m' \setminus m)$$

then

$$\rho_{m,m'} = \begin{cases} c & \text{if } |m \Delta m'| = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Calculate c to ensure $\sum_{m'} \rho_{m,m'} = 1$ for every $m \in \mathcal{M}$.

- (ii) Construct the dimension-changing updates of a reversible jump MCMC algorithm targeting $\pi(\beta_m, \sigma, m | y)$, including an update adding a component to β_m and an update deleting a component from β_m , with acceptance probabilities given in terms of the model elements. Use ρ to propose the new model.

2. (a) [9 marks] Let $X_1, X_2, X_3, \dots$ be an infinite exchangeable sequence of binary random variables. For $n = 1, 2, 3, \dots$ and $x_i \in \{0, 1\}$, $i = 1, \dots, n$ let

$$\Pr\{X_1 = x_1, \dots, X_n = x_n\} = p(x_1, \dots, x_n)$$

give the Probability Mass Function (PMF) for $X_1, \dots, X_n$.

- (i) What does it mean to say that the sequence is an infinite exchangeable sequence?
(ii) Show that there exists a generative model for $X_1, X_2, X_3, \dots$ given in terms of a cumulative distribution function $F : [0, 1] \rightarrow [0, 1]$, in which

$$Q \sim F \text{ and, independently for } i = 1, 2, 3, \dots, X_i \sim \text{Bernoulli}(Q).$$

You may quote de Finetti's theorem without proof.

- (iii) Suppose that, for some fixed $0 < a < 1$ and $k(x) = \sum_{i=1}^n x_i$, we set

$$p(x_1, \dots, x_n) = \frac{a^{k(x)}(1-a)^{n-k(x)}}{3} + \frac{4}{3} \frac{\Gamma(k(x)+2)\Gamma(n-k(x)+1)}{\Gamma(n+3)}.$$

Give an explicit generative model for the infinite exchangeable sequence $X_1, X_2, X_3, \dots$ admitting $p(x_1, \dots, x_n)$ as its marginal PMF for every $n \geq 1$.

- (b) [11 marks] Let $G = (V, E)$ be a directed graph with nodes $V = \{1, \dots, n\}$ and edges $E \subseteq V \times V$. A Directed Acyclic Graph (DAG) is a directed graph with no directed loops. A walk following the directed edges cannot return to its starting node.

For $n = 1, 2, 3, \dots$ let $\mathcal{G}_n$ be the set of all distinct DAG's on n labeled nodes and let $|\mathcal{G}_n|$ give the number of such DAG's. There are three DAG's on $n = 2$ nodes with $V = \{1, 2\}$ and $G_i = (V, E_i)$, $i = 1, 2, 3$. Possible edge sets are $E_1 = \emptyset$, $E_2 = \{(1, 2)\}$ and $E_3 = \{(2, 1)\}$, so $|\mathcal{G}_2| = 3$. There are $|\mathcal{G}_3| = 25$ DAG's on $n = 3$ nodes.

- (i) Let $\pi_{U,n}(g)$, $g \in \mathcal{G}_n$ give the uniform distribution on DAG's with n nodes, so that

$$\pi_{U,n}(g) = \frac{\mathbb{I}_{g \in \mathcal{G}_n}}{|\mathcal{G}_n|}.$$

For $G \sim \pi_{U,3}$ and $G = (V, E)$ let $G_{-3} \in \mathcal{G}_2$ be the DAG obtained from G by removing node 3 and any edge $(i, 3) \in E$ or $(3, i) \in E$, $i = 1, 2$ connected to node 3. Show that G_{-3} is not distributed according to $\pi_{U,2}$.

- (ii) Let $\mathcal{S}$ be the set of all subsets of $\mathcal{G}_2 \cup \mathcal{G}_3$. For $G \in \mathcal{G}_3$ and $A \in \mathcal{S}$ event A occurs if $G \in A$ or $G_{-3} \in A$. Explain briefly why there does not exist a probability distribution over $\mathcal{G}_3$ expressing prior preferences over sets $A \in \mathcal{S}$ which match the prior preferences expressed by $\pi_{U,n}$, $n = 2, 3$.

- (iii) Suppose for g a graph on n nodes we have a function $\text{is.DAG}(g)$,

$$\text{is.DAG}(g) = \begin{cases} 1 & \text{if } g \in \mathcal{G}_n \\ 0 & \text{otherwise,} \end{cases}$$

so $\text{is.DAG}(g)$ checks g is a DAG. Give an MCMC algorithm targeting $\pi_{U,n}$ for fixed $n \geq 2$. Explain briefly why your MCMC chain is ergodic.